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THE EFFECTS OF SLEEP RESTRICTION ON FOOD INTAKE: THE IMPORTANCE OF INDIVIDUAL CHARACTERISTICS

Brunet J¹, McNeil J², Jaeger Hintze L³, Blais C¹, Doucet E³, Forest G¹¹Université du Québec en Outaouais, Gatineau, QC, CANADA,²Alberta Health Services, Calgary, AB, CANADA, ³University of Ottawa, Ottawa, ON, CANADA

Introduction: Acute sleep restriction (SR) often leads to an increase in energy intake (EI). However, large variability in food intake is often observed, which suggests that individual factors may affect EI after SR. The objective of this study was to explore the influence of personality traits (excitement seeking, impulsiveness and sense of competence), implicit attitudes toward food, and sensitivity to reward and punishment on EI after sleep loss.

Methods: Seventeen subjects (11men; 18-33y) completed a personality inventory (NEO-PI-3), an Implicit Association Test (IAT; this test measures implicit attitudes toward healthy and unhealthy foods), and the Sensitivity to Punishment and Sensitivity to Reward Questionnaire (SPSRQ). *Ad libitum* EI over 24h was assessed following an habitual sleep night and a 50% SR held during the second half of the night. The difference in EI between sleep conditions (Δ EI) was calculated for each subject. Correlations between Δ EI and subscales of the NEO-PI-3, IAT score and SPSRQ scores were computed. A multiple linear regression model was performed to examine the unique contribution of each variable that was significantly associated with Δ EI. We controlled for gender, resting energy expenditure and fat% in these analyses.

Results: Δ EI was associated with excitement seeking ($r=-0.70$, $p<0.01$) and IAT score ($r=0.70$, $p<0.01$), suggesting that less excitement seeking and more positive attitudes towards unhealthy food lead to greater EI after SR. Impulsiveness, sense of competence and SPSRQ scores were not associated with Δ EI. Multiple linear regression modeling demonstrated that excitement seeking and implicit attitudes collectively explain 62% of the variance in EI after SR ($p<0.001$). Excitement seeking was the largest predictor ($\beta=-0.63$, $p<0.01$), followed by implicit attitudes ($\beta=0.40$, $p<0.05$).

Conclusion: These results suggest that excitement seeking and implicit attitudes toward food are key factors in explaining variations in EI after sleep loss. Since implicit attitudes are strongly related to educational and personal beliefs/values and personality traits are shaped by early life experiences which tend to remain stable over time, addressing weight gain issues in sleep restriction settings (e.g. rotating shifts) may be more challenging than simply addressing eating behaviors.

Support (If Any):

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CHARACTERIZING SLEEP, CIRCADIAN RHYTHMS, AND EYE CLOSURE IN ACOMYS CAHIRINUS (CAIRO SPINY MOUSE)

Wang C¹, Guerriero LE¹, Ajwad AA², Huffman DM², Sunderam S²,Brooks TC¹, Seifert AW¹, O'Hara BF¹¹Department of Biology, University of Kentucky, Lexington, KY,²Department of Biomedical Engineering, University of Kentucky, Lexington, KY

Introduction: In order to better understand the functions and origins of sleep, sleep should be studied across a variety of species. We aim to characterize sleep and wake in *Acomys cahirinus*, the Cairo spiny mouse. Few studies on the circadian activity of this species are available and nothing is known of their sleep behavior. Therefore, we have

begun to characterize sleep, circadian rhythms, and eye closure for this species (*A. cahirinus*) in greater detail and alongside the well-studied house mouse (*Mus musculus*).

Methods: Sleep and wake states were determined using a piezoelectric system for individually housed mice for 7 days under 12:12 LD condition. Four infrared cameras were set up around the cage to monitor activity patterns of *A. cahirinus* in light and dark conditions. Starting with one mouse, two more were added to the cage every four days ending with a total of five to study the effect of group housing on activity. Then, mice were surgically instrumented for tethered electroencephalogram (EEG) recording. To research eye closure of *A. cahirinus*, we set up a light flashing experiment to challenge the eye.

Results: We found *A. cahirinus* and *M. musculus* to be primarily nocturnal, but with distinct behavioral patterns. The activity of *A. cahirinus* sharply increases at dark onset, but surprisingly, decreases sharply just one hour later. *A. cahirinus* is more active in the first half of the night than the second half in both single and group housing. Based on EEG analysis, *A. cahirinus* sleeps more than *Mus* during both day and night. The proportion of REM is significantly higher (nearly tripled). The proportion of wakefulness is more in the first half of the night. We also found that *A. cahirinus* do not close their eyes during sleep periods of the day or night, even with lights flashing.

Conclusion: *A. cahirinus* has different sleep and circadian behavior than the standard laboratory nocturnal mouse. They sleep more than *Mus* during both daytime and night time. REM percentage is significantly higher.

Support (If Any): NSF and OISE (IOS-1353713), NIH grant NS083218, Kentucky Cabinet for Economic Development.

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DIRECT ACTIVATION OF G-PROTEIN-COUPLED INWARD RECTIFYING K⁺ CHANNELS PROMOTE SLEEP IN RODENTS

Xie X¹, Zou B¹, Cao W¹, Pascual C¹, Xiao K¹, Guan Z², Lindsley C³, Weaver C³, Fang J²¹AfaSci Research Laboratories, Redwood City, CA, ²Department of Psychiatry, Pennsylvania State University College of Medicine, Hershey, PA, ³Department of Pharmacology, Vanderbilt University School of Medicine, Nashville, TN

Introduction: The most common type of sleep disorder is insomnia which is characterized by difficulty falling asleep and/or maintaining sleep. Chronic insomniacs respond poorly to current treatments including benzodiazepines and non-benzodiazepines. Activation of G-protein-gated inward rectifying K channels (GIRKs) by GABA_B agonists baclofen or γ -hydroxybutyric acid (GHB) predominantly promote non-rapid eye movement (NREM) sleep in rodents and humans. GHB especially reduces sleep fragmentation in narcoleptic patients. The present study explored the link between direct GIRK activation and sleep regulation in rodents using our potent and selective GIRK channel activator ML297.

Methods: Whole-cell patch clamp recordings from hypocretin/orexin-EGFP neurons and extracellular recordings of mouse hippocampal CA1 area in mouse brain slices were made to study cellular and synaptic actions of ML297. Wake activity and locomotion were measured using the noninvasive behavior monitoring system SmartCageTM. Furthermore, wakefulness, REM and NREM sleep were determined using the electroencephalogram/electromyogram (EEG/EMG) in wild type mice (C57BL/6) implanted with electrodes.

Results: Bath-application of ML297 (5 μ M) hyperpolarized resting potential, decreased membrane input resistance and blocked